8 PUZZLE PROBLEM

```
PS D:\1BM24CS318\LAB 2> py .\8puzzledfs.py
Initial Board Setup:
 1 2 3
 4 0 5
 7 8 6
-----------

--- Running Depth First Search (DFS) ---
DFS found a solution path in 24 moves.
 1 2 3
 4 0 5
 7 8 6
-----------
Action taken: Move blank tile 'Up'
 1 0 3
 4 2 5
 7 8 6
-----------
Action taken: Move blank tile 'Right'
 1 3 0
 4 2 5
 7 8 6
-----------
Action taken: Move blank tile 'Down'
 1 3 5
 4 2 0
 7 8 6
-----------
Action taken: Move blank tile 'Down'
 1 3 5
 4 2 6
 7 8 0
-----------
Action taken: Move blank tile 'Left'
 1 3 5
 4 2 6
 7 0 8
-----------
Action taken: Move blank tile 'Up'
 1 3 5
 4 0 6
 7 2 8
-----------
Action taken: Move blank tile 'Up'
 1 0 5
 4 3 6
 7 2 8
-----------
Action taken: Move blank tile 'Left'
 0 1 5
 4 3 6
 7 2 8
```

```
Action taken: Move blank tile 'Down'
 4 1 5
 0 3 6
 7 2 8
----------
Action taken: Move blank tile 'Right'
 4 1 5
 3 0 6
 7 2 8
----------
Action taken: Move blank tile 'Down'
 4 1 5
 3 2 6
 7 0 8
----------
Action taken: Move blank tile 'Right'
 4 1 5
 3 2 6
 7 8 0
----------
Action taken: Move blank tile 'Up'
 4 1 5
 3 2 0
 7 8 6
----------
Action taken: Move blank tile 'Left'
 4 1 5
 3 0 2
 7 8 6
----------
Action taken: Move blank tile 'Left'
 4 1 5
 0 3 2
 7 8 6
----------
Action taken: Move blank tile 'Up'
 0 1 5
 4 3 2
 7 8 6
----------
Action taken: Move blank tile 'Right'
 1 0 5
 4 3 2
 7 8 6
----------
Action taken: Move blank tile 'Right'
 1 5 0
 4 3 2
 7 8 6
----------
```

```
-----------
Action taken: Move blank tile 'Down'
 1 5 2
 4 3 0
 7 8 6
-----------
Action taken: Move blank tile 'Left'
 1 5 2
 4 0 3
 7 8 6
-----------
Action taken: Move blank tile 'Up'
 1 0 2
 4 5 3
 7 8 6
-----------
Action taken: Move blank tile 'Right'
 1 2 0
 4 5 3
 7 8 6
-----------
Action taken: Move blank tile 'Down'
 1 2 3
 4 5 0
 7 8 6
-----------
Action taken: Move blank tile 'Down'
 1 2 3
 4 5 6
 7 8 0
-----------
```

8 PUZZLE IDS

```
PS D:\1BM24CS318\LAB 2> py .\8puzzleIdf.py
=== IDS ===
  IDS depth limit  0 -> nodes so far: 1
  IDS depth limit  1 -> nodes so far: 5
  IDS depth limit  2 -> nodes so far: 14
  IDS depth limit  3 -> nodes so far: 30
Solution length: 3 moves, total nodes visited: 30
Moves: ['Right', 'Down', 'Right']

Initial state:
1 2 3
_ 4 6
7 5 8

Step 1: move blank Right
1 2 3
4 _ 6
7 5 8

Step 2: move blank Down
1 2 3
4 5 6
7 _ 8

Step 3: move blank Right
1 2 3
4 5 6
7 8 _
```